Incident report INC-2025-031: reactor temperature overshoot on TIC-1005

Unit: U-200 Hydrotreating · Equipment: R-1005 Reactor · Date of event: 2025-07-18 · Report issued: 2025-07-30 · Classification: process upset, no release, no injury

Summary

During a feed change on 2025-07-18 at 14:20, the reactor temperature TIC-1005 overshot its setpoint of 62 °C by 7 °C and reached the high alarm (68 °C). The temperature returned to setpoint after 25 minutes. Since this event the operators switch TIC-1005 to manual before feed changes and often leave it in manual for the rest of the shift.

Causes

1. The feed change was made in one step of 10 m^3/h instead of steps of 5 m^3/h (DOC-U200-OM-001, section 4).
2. The integral action of TIC-1005 was too fast for the process (time constant about 310 s, dead time about 35 s). The integral wound up during the response and caused the overshoot.
3. The recycle flow FIC-1003 was cycling at the same time because of the stiction of FV-1003 and added a disturbance to the reactor.

Actions

Action	Owner	Due	Status
Retune TIC-1005 with slower integral action (integral gain 0.15)	Process control engineering	2025-08-15	done
Repeat the feed-change training for all shifts	Operations	2025-09-30	done
Return TIC-1005 to automatic during feed changes, with a target of 95 % of the time in automatic	Operations and process control	2025-10-31	open: the loop is still kept in manual during feed changes

Lessons

When operators lose trust in a loop, retuning alone does not bring it back to automatic. The return to automatic needs to be agreed with the shifts and watched.